

Learning to Herd Under Limited Visibility: Recurrent PPO with Procedural Curricula for Single-Agent Shepherd

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Abstract—Shepherding asks one agent to move a flock of reactive agents into a goal region. Collect-and-drive heuristics solve open fields well but degrade when sensing is local, obstacles are present, or the flock splits. We train a recurrent PPO herder with limited visibility on procedurally generated obstacle layouts, with domain randomization, a geometric shaped reward, and an adaptive curriculum, and compare it with a collect-and-drive heuristic and a behavioral clone on 150 held-out episodes per scenario over three training seeds. A policy trained on fixed layouts had succeeded in no held-out split-field or open-field episode; the procedurally trained policy succeeds in 28% and 21% of them on average across seeds. On the split field, two of three seeds deliver more of the flock than the heuristic (58% and 51% versus 32%), but the third delivers 18%. The heuristic stays ahead on the procedural test suite (47% versus 30%) and the open field, and no agent solves corridor or narrow-gate maps. Procedural test scores agree across seeds within 7 points, while single-map scores vary by up to 39 points. We also document the curriculum failure modes found in training and their fixes.

Index Terms—Shepherding, Reinforcement learning, Partial observability, Curriculum learning, Domain randomization, Generalization

I. INTRODUCTION

Shepherding is the problem of steering a group of passive, interacting agents into a target region using one or a few active agents [1], [2]. The herder never carries a sheep. It can only change where the flock wants to go by choosing where to stand. The same structure appears in livestock robotics [3], crowd guidance, and the indirect control of robot swarms [4], [5].

Strömbom *et al.* showed that a short program reproduces the GPS traces of working sheepdogs: collect the furthest sheep when the flock is too spread out, otherwise drive the flock from behind toward the goal [2]. That rule assumes the dog can see the whole flock and that the field is mostly open. Both assumptions fail when visibility is limited, when obstacles block the direct route, or when the flock splits into separate groups. Learning a policy avoids tuning the rule for each geometry, but the learning problem is a partially observable Markov decision process (POMDP) with a long horizon, many interacting bodies, and sparse success.

Our first attempt illustrates the main risk. A recurrent PPO policy trained on four fixed obstacle layouts, with the

flock always spawned opposite the goal, succeeded in 75% of training episodes but in none of the held-out split-field or open-field episodes, where the heuristic succeeded in 25% and 100% (small evaluation budget, single seed). The policy had learned the layouts rather than how to herd, a failure mode well documented for RL benchmarks with a fixed set of levels [6]. This paper describes the redesign that followed and measures how far it goes.

Our contributions are:

- A herding environment with procedurally generated obstacle topologies, randomized flock size, dynamics, sensing noise and goals, and a held-out test distribution with wider parameter ranges than training (Section III).
- A recurrent PPO agent with an egocentric observation, a geometric shaped reward, and an adaptive curriculum. We document four failure modes of the curriculum and checkpointing pipeline and the fixes for each (Section IV, Section VI.C).
- A comparison over three training seeds with a collect-and-drive heuristic and a behavioral-cloning baseline, on 900 held-out episodes per agent and seed, with bootstrap confidence intervals (Section VI).

The learned policy transfers to held-out geometry that defeated the earlier version, and in two of three seeds it beats the heuristic when bars split the flock. It does not match the heuristic on the procedural test suite, and the seed-to-seed spread on single maps is large enough that one run would have supported a much stronger claim.

II. RELATED WORK

Heuristic shepherding. Reynolds’ boids established local attraction, alignment and repulsion as a model of flocking [7]. Lien *et al.* described geometric herding primitives for motion planning [1], and Vaughan *et al.* herded real ducks with a mobile robot [4]. The collect-and-drive algorithm of Strömbom *et al.* [2] and its robotic extensions [8] are the standard single-herder baseline. Pierson and Schwager analyze herding of non-cooperative agents from a control perspective [5], and King *et al.* review robot herding inspired by biology [3].

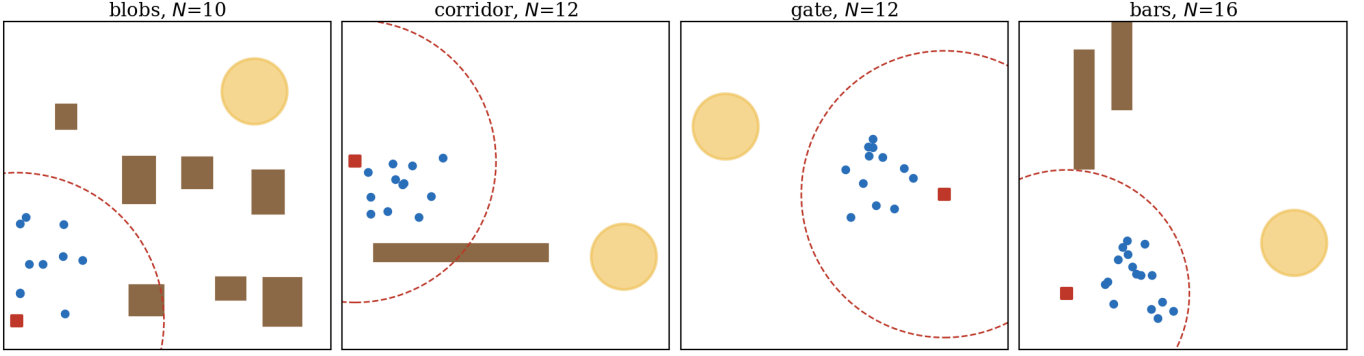


Fig. 1. Samples from the held-out procedural test distribution, one per obstacle topology. Red square: dog. Dashed circle: its visibility radius. Blue: sheep. Yellow disc: goal region. N is the flock size.

Learning to shepherd. Hussein *et al.* learn the collect and drive behaviors with curriculum PPO instead of a waypoint generator [9]. Hasan *et al.* treat cooperative multi-shepherd herding as payload protection [10], and Napolitano *et al.* learn decentralized policies for several herders and non-cohesive targets [11]. These works usually assume wider sensing than ours, or split the task into target assignment and single-target driving.

Generalization and evaluation. Recurrent policies are the usual response to partial observability [12], [13]. Domain randomization [14] and curricula [15] widen the training distribution, and procedurally generated levels with a held-out test set are the standard way to measure whether a policy generalizes [6]. Agarwal *et al.* show that RL results from few runs and few episodes are often not reliable [16]. We therefore report bootstrap intervals [17] and flag single-seed results. Behavioral cloning [18], [19] is a common imitation baseline, and its compounding error under distribution shift is well known [20].

III. ENVIRONMENT

A. Workspace and flock

The workspace is the square $[0, 20]^2$. A dog at position d_t herds N sheep s_t^i toward a goal g among at most eight axis-aligned rectangular obstacles. The action $a_t \in [-1, 1]^2$ is normalized to unit length, so the dog always moves one unit per step, and is clipped to the longest collision-free prefix of the step. An episode succeeds when every sheep is within $\rho = 2$ of g and is truncated after 700 steps.

Each sheep moves at a constant speed v_s in the direction of a sum of forces, following [2], [7]. It flees the dog with unit weight when the dog is within the flee radius R_f and is pulled toward the flock centroid with gain α . It is repelled by neighbors closer than a distance R_r and pulled toward a leader, the sheep nearest the centroid, with gain β . A linear repulsion pushes it away from obstacles within a threshold distance, and Gaussian noise with standard deviation 0.02 is added. While a sheep is inside the flee radius it also receives a small pull of 0.03 toward the goal. The nominal values are $v_s = 0.32$, $R_f = 5.5$, $\alpha = 0.07$, $R_r = 1.0$ and $\beta = 0.04$.

B. Observation

The dog sees sheep within a Euclidean visibility radius R_v (nominally 7.5). Obstacles do not block sight. The policy receives a 98-dimensional egocentric vector with no absolute position. It contains the direction and distance to the goal, the distances to the arena walls, the previous action, and summary statistics of the visible sheep (fraction visible, centroid offset, mean and maximum distance, spread). A memory block holds the offset to the last seen flock centroid, the time since it was seen, and its direction and distance to the goal, followed by the remaining time. Next come 16 sheep slots of (relative x , relative y , visible flag), with invisible sheep and unused slots set to zero, and 24 lidar ranges to obstacles and walls, up to 15 units. Sheep positions carry Gaussian observation noise, and executed actions carry Gaussian action noise. The fixed slot count lets one network handle flocks of 6 to 16 sheep.

C. Scenarios and splits

Layouts are drawn from five topologies: *open* (no obstacles), *blobs* (scattered rectangles), *corridor* (parallel walls), *gate* (a wall across the field with one opening) and *bars* (short barriers that split the flock). Fig. 1 shows one sample of each obstacle topology. Goals are uniform in $[2.4, 17.6]^2$, and the flock spawns in a small box at least 9 units from the goal with the dog behind it. Obstacles near the spawn or the goal are removed, and a flood fill rejects layouts in which the goal is unreachable.

We use three splits. *Train* samples topologies, dynamics and noise from the training ranges, scaled by the curriculum (Section IV.C). *Validation* uses the same ranges at full breadth with a separate seed stream, and selects the checkpoint. *Test* (test_procedural) uses wider ranges that extend past the training ranges at both ends, on seeds not used in training or validation. Its flocks have 8 to 16 sheep instead of 6 to 14. It also has more obstacle blobs, narrower gates (2.2 to 4.0 units instead of 3.0 to 6.0), longer walls, and visibility radii from 4.5 to 10.0 instead of 5.0 to 9.5. The test ranges for sheep speed, cohesion, repulsion, flee radius and noise are wider in the same way.

Five hand-authored presets with fixed parameters and 10 sheep serve as further held-out maps, none used for training or

model selection. In *corridor* the flock must pass two staggered pairs of walls, *dense* has five larger blobs near the start, *narrow gate* has two offset walls with openings, *split field* has three horizontal bars between the spawn and the goal, and *open field* has no obstacles, a small visibility radius (5.6) and faster sheep.

IV. METHOD

A. Reward

The per-step reward is a sum of geometric terms. Let c_t be the flock centroid, $D = 20\sqrt{2}$ the workspace diagonal, H_t the area of the flock’s convex hull, δ_t and m_t the mean and maximum sheep-to-goal distance, and ν_t the visible fraction. The base term rewards a centroid near the goal and a compact flock, penalizes a dog standing between the flock and the goal, and rewards staying near the flock:

$$r_t^{\text{base}} = \left(1 - \frac{\|c_t - g\|}{D}\right) - 0.1 \frac{H_t}{400} - 0.5\mathbb{1}[\|d_t - g\| < \|c_t - g\|] + 0.3 \left(1 - \frac{\|d_t - c_t\|}{D}\right)^{(1)}$$

The full reward adds progress, visibility, flock-cohesion, positioning and collision terms:

$$r_t = r_t^{\text{base}} + 3.0(\delta_{t-1} - \delta_t) + 2.5(m_{t-1} - m_t) + 0.8(\nu_t - \nu_{t-1}) - 0.05(1 - \nu_t) - 0.25\mathbb{1}[\nu_t = 0] - 0.05n_t^{\text{stray}} + 0.35\left(1 - \frac{\|d_t - p_t\|}{D}\right) + r_t^{\text{coll}}, \quad (2)$$

where n_t^{stray} counts sheep more than 1.8ρ from the centroid. The drive point p_t lies behind the centroid on the goal axis at distance $\min(0.7R_v, 0.9R_f)$, which is where a collect-and-drive dog would stand [2]. The collision term is -0.2 when the dog first touches an obstacle or wall and -0.03 per step while contact lasts. Success adds a terminal bonus of $+20$. We lowered this bonus from an earlier $+125$ because, with reward normalization, a bonus that large dominated the value targets. The progress terms are differences of potentials [21].

B. Recurrent PPO

The policy is recurrent PPO [22] with an LSTM actor-critic of hidden size 256 (MlpLstmPolicy from sb3-contrib [23]). We collect 256 steps from each of 12 parallel environments per update, then train for 10 epochs with batch size 384, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip range 0.2 and entropy coefficient 0.005. The learning rate anneals linearly from 3×10^{-4} to zero over 6×10^5 steps. Observations and rewards are normalized with running statistics. Every 25k steps the policy is evaluated on 30 validation episodes, and the checkpoint with the highest mean fraction of the flock at the goal is kept together with its normalization statistics. Training takes about 75 minutes on a 16-core CPU.

C. Adaptive curriculum

The curriculum has four stages, $\sigma \in \{0, 0.33, 0.66, 1\}$. Stage 0 samples open, blob and bar layouts, and shrinks every randomized range to 25% of its width around the nominal

value. Stage 0.33 adds gates and widens the ranges to 60%. Stage 0.66 adds corridors, uses the full ranges, and randomizes the flock size. Stage 1 uses the same distribution as stage 0.66 but has stricter gates. The goal is random at every stage.

A callback tracks a window of the last 40 training episodes. It promotes the agent when the window meets the next stage’s thresholds on the fraction of the flock at the goal, visibility, collision events and progress reward, after a minimum share of the training budget. The thresholds for stages 0.33, 0.66 and 1 require at least 30%, 55% and 70% of the flock at the goal, and at most 22, 19 and 16 collision events per episode. Two rules prevent oscillation. After any stage change, the stage is fixed for 25k steps. A demotion also requires missing a threshold by 25% of its value. Collision counts can block a promotion but never cause a demotion; Section VI.C explains why.

D. Baselines

Heuristic. The expert is a collect-and-drive controller in the style of [2]. When at least four sheep are visible and the two furthest apart are more than 2.0 units apart, it splits the visible sheep into two clusters and works on the one further from the goal. If a sheep in that cluster is more than 3.2 units from the cluster centroid, the dog moves behind that sheep to collect it. Otherwise it moves to a drive point behind the centroid. An obstacle-avoidance term is added to the command. With no sheep visible, it returns to the last seen centroid or searches toward the goal. The heuristic has an advantage over the learned agents: it reads the absolute dog position and every obstacle rectangle regardless of visibility. It also uses the nominal visibility and flee radii rather than the randomized ones.

Behavioral cloning (BC). A random forest [24] with 300 trees and depth 18 regresses the expert’s action from raw coordinates plus 15 geometric summaries of the visible flock. On an episode-level hold-out it reaches R^2 of 0.59 and 0.69 on the two action components, with a mean heading error of 29° . The most important feature is the fraction of visible sheep in the focus cluster (impurity importance 0.147), which is the quantity the expert uses to detect a split. The forest was trained on demonstrations from an earlier version of the environment, in which goals were confined to the far quadrant. Those demonstrations are no longer available, so we could not retrain it on the procedural distribution. Its results below therefore test a clone trained on a narrower distribution, not cloning in general.

V. EXPERIMENTAL SETUP

Each agent is evaluated on the procedural test split and the five presets, 150 episodes per scenario for the heuristic and recurrent PPO and 100 for BC. All agents use deterministic actions and see the same episodes: seeds start at 100000 for every agent, so layouts, goals and dynamics match. The heuristic and BC read the older observation format, which has exactly ten sheep slots. For a matched comparison we therefore also fix the flock at ten sheep for recurrent PPO; we report the randomized-flock result separately.

TABLE I

HELD-OUT RESULTS WITH A 10-SHEEP FLOCK; ALL AGENTS SEE THE SAME EPISODES. BASELINES: FRACGOAL WITH 95% BOOTSTRAP INTERVAL. RECURRENT PPO: MEAN OVER THREE SEEDS, WITH THE LOWEST AND HIGHEST SEED IN PARENTHESES; SR AND DIST. ARE SEED MEANS. BOLD MARKS THE BEST FRACGOAL WHEN IT IS AHEAD OF EVERY SEED OF EVERY OTHER AGENT. WITH THE RANDOMIZED 8 TO 16 SHEEP FLOCK, RECURRENT PPO SCORES 0.234 (0.209, 0.281), SR 0.13, ON THE PROCEDURAL TEST.

Scenario	Heuristic			Behavioral cloning			Recurrent PPO (ours)		
	FracGoal	SR	Dist.	FracGoal	SR	Dist.	FracGoal	SR	Dist.
Procedural test	0.468 [0.395, 0.537]	0.35	3.44	0.010 [0.001, 0.024]	0.00	12.20	0.296 (0.261, 0.326)	0.20	4.47
Split field	0.317 [0.276, 0.360]	0.08	3.97	0.196 [0.161, 0.233]	0.01	3.86	0.424 (0.184, 0.576)	0.28	3.19
Dense	0.269 [0.207, 0.333]	0.19	6.41	0.062 [0.036, 0.090]	0.00	7.83	0.205 (0.040, 0.289)	0.06	4.55
Open field	0.989 [0.975, 1.000]	0.98	1.21	0.599 [0.519, 0.674]	0.39	2.17	0.370 (0.233, 0.565)	0.21	3.02
Corridor	0.000 [0.000, 0.000]	0.00	12.21	0.008 [0.000, 0.023]	0.00	10.27	0.004 (0.000, 0.007)	0.00	11.92
Narrow gate	0.003 [0.000, 0.007]	0.00	6.75	0.021 [0.008, 0.036]	0.00	5.36	0.005 (0.000, 0.011)	0.00	10.12

The main metric is the fraction of the flock inside the goal region at the end of the episode (FracGoal). It gives partial credit, which matters when most episodes end with a few stragglers outside the goal. We also report success rate (SR, all sheep inside) and the mean sheep-to-goal distance. The 95% intervals for FracGoal are percentile bootstrap intervals over episodes with 2000 resamples [17]. We train recurrent PPO with three seeds (0, 1, 2) and the same configuration, and evaluate the checkpoint that each run selected on validation. For recurrent PPO we report the mean over seeds and, in parentheses, the lowest and highest per-seed value.

VI. RESULTS

A. Held-out comparison

Table I and Fig. 2 give the results. On the procedural test split the heuristic delivers 47% of the flock and succeeds in 35% of episodes. Recurrent PPO delivers 30% on average and succeeds in 20%. All three seeds fall between 26% and 33%, and each seed’s interval lies below the heuristic’s. With its randomized flock sizes the test is harder still, and FracGoal drops to 23%.

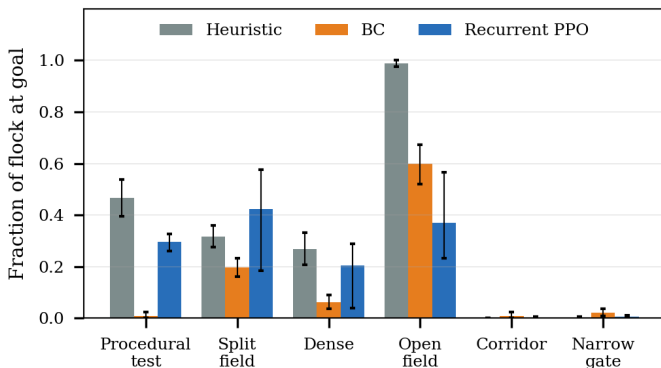


Fig. 2. Fraction of the flock at the goal on the held-out scenarios. Baseline error bars are 95% bootstrap intervals over episodes. The recurrent PPO bar is the mean of three seeds, and its error bar spans the lowest and highest seed.

The presets give a more varied picture, and the seeds disagree much more on them (Table II). On the split field, seeds 0 and 1 deliver 58% and 51% of the flock, and both

intervals lie above the heuristic’s 32%. Seed 0 also succeeds in 45% of episodes against the heuristic’s 8%. The bars tend to break the flock into groups on different sides of an obstacle, and the heuristic works on one cluster at a time. In 92% of its split-field episodes it runs out of time before the whole flock reaches the goal (mean episode length 660 of 700 steps). Seed 2, however, delivers only 18% on this map. On the dense map seeds 0 and 1 match the heuristic’s delivery within noise and leave the flock closer to the goal, while seed 2 delivers 4%. On the open field the heuristic is nearly perfect (99%, finishing in 191 steps on average). The seeds deliver between 23% and 57%. This is the setting that Strömbom’s rule was designed for, and no seed recovers it fully. Corridor and narrow-gate maps defeat all three agents. Both require the dog to push the whole flock through an opening, and neither the reward nor the heuristic has a term for that.

TABLE II

FRACGOAL PER RECURRENT PPO SEED ON THE HELD-OUT SCENARIOS (150 EPISODES EACH), AND EACH SEED’S BEST VALIDATION SCORE AND THE STEP AT WHICH IT WAS REACHED.

Scenario	Seed 0	Seed 1	Seed 2
Procedural test	0.261	0.302	0.326
Split field	0.576	0.511	0.184
Dense	0.289	0.286	0.040
Open field	0.565	0.311	0.233
Corridor	0.007	0.005	0.000
Narrow gate	0.005	0.000	0.011
Best validation	0.410 @ 600k	0.455 @ 600k	0.378 @ 175k

On the procedural test, which averages over many layouts, the three seeds lie within 7 points of each other; on single maps they differ by up to 39 points. Seed 2 scores best on the procedural test but worst on three presets. Its selected checkpoint comes from 175k steps, early in training, so validation on the procedural distribution picked a policy that does well on average but fails on particular layouts. Had we trained only seed 0, we would have reported 45% and 49% success on the split and open fields; the three-seed means are 28% and 21%.

Compared with the earlier structured policy, which succeeded in none of the held-out split-field and open-field episodes, every seed of the procedurally trained policy succeeds in some of them (3% to 49%). That earlier result used a small evaluation budget, so the comparison is qualitative.

The clone does worst in nearly every scenario. It keeps some competence on the open field (60% delivered), the setting closest to its training data, and falls to 1% on the procedural test. Its 29° offline heading error compounds in closed loop [20], and the change in goal distribution puts most test states outside its training data.

B. Training dynamics

Fig. 3 shows validation performance and the curriculum stage for the three seeds. Seeds 0 and 1 improve through the second half of training as the learning rate decays, and their final checkpoints are also their best (0.41 and 0.46). Seed 2 reaches 0.38 at 175k steps and stays between 0.21 and 0.36 afterwards. In every seed, collision events per training episode fall to about 14, although collisions are never a reason for demotion.

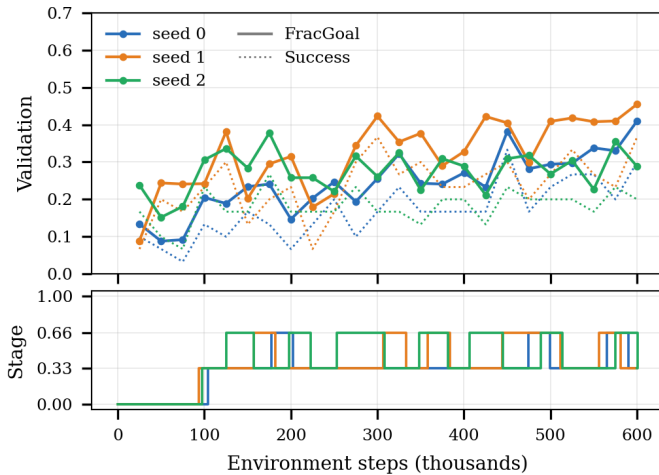


Fig. 3. Validation FracGoal (solid) and success rate (dotted) every 25k steps, 30 episodes each, and the curriculum stage (bottom), for three seeds.

All seeds reached stage 0.33 at 94k to 104k steps and stage 0.66 by 178k, then alternated between the two, and none reached stage 1. Each demotion from 0.66 came from delivery: at full breadth the agent could not keep FracGoal above the demotion bound of 0.41. The share of training at stage 0.66 was 13%, 28% and 39% for seeds 0, 1 and 2, so most learning happened at 60% breadth, without corridors or randomized flock sizes. That is consistent with the weak corridor results and with the gap between the fixed and randomized flock results on the test split. More time at full breadth did not by itself help, though: seed 2 spent the most time there and has the flattest validation curve.

C. Training stability

Four earlier runs failed or were stopped, each for a different reason (Table III). In the first, stage 0 pinned the goal to one corner, and validation FracGoal, which always uses a random goal, was zero at all 17 checkpoints. Next, the collision

threshold became the only binding constraint, because arena walls count as collisions and random goals often lie near a wall. Each promotion then added harder layouts, which raised collisions enough to demote the agent while delivery was still improving. Once collisions no longer caused demotions, they fell below the old thresholds on their own, because the reward still penalizes them. Finally, the best checkpoint had been saved without its observation-normalization statistics, and with 12 validation episodes (standard error about 0.1) checkpoint selection mostly picked the luckiest draw.

TABLE III
FAILURE MODES FOUND IN SUCCESSIVE TRAINING RUNS AND THE CHANGE THAT REMOVED EACH.

Run	Symptom	Fix
1	Validation FracGoal 0 for 433k steps; stage flips every rollout	Random goal from stage 0; 25k-step stage lock and 25% demotion margin
2	Stuck at stage 0.33 by wall contacts	Collision thresholds 14/11/9 raised to 22/19/16
3	Demoted after each promotion as collisions rise	Collisions gate promotion only
4	Best checkpoint lacks matching normalization; noisy selection; late drift	Save statistics with checkpoint; 30 validation episodes; linear learning-rate decay

Runs 2 and 3 differed only in thresholds that did not bind before 75k steps, yet reached training FracGoal of 0.52 and 0.30 at that point, because parallel environments are not bit-reproducible.

VII. DISCUSSION

First, a narrow training distribution was the main cause of the earlier collapse on held-out maps. With procedural layouts, random goals and randomized dynamics, every seed of the same algorithm transfers to some degree to maps it never saw. Second, a learned policy can do better than the hand-written rule when obstacles split the flock, but training does not produce such a policy reliably: two of three seeds do, and one does not. We have not yet analyzed the learned trajectories, but the memory features and LSTM state give the policy a way to track sheep it can no longer see, which the heuristic lacks beyond the last seen centroid. The seed spread also bears on evaluation practice. Scores on single hand-designed maps changed the conclusion from seed to seed, while the procedural test did not, which supports reporting averages over many sampled layouts and several seeds [6], [16].

The open question is whether a learned policy can match the heuristic across the whole test distribution. Three factors likely hold it back: the curriculum rarely trained at full difficulty, the heuristic reads the full obstacle map while the policy has 24 lidar rays, and the reward has no term for pushing a flock through an opening.

Several changes follow directly. The demotion bound at stage 0.66 could be relaxed so that most training happens at full breadth, although seed 2 suggests that this alone is not enough. Selecting checkpoints on a validation set that includes

more layout types, or averaging several checkpoints, could reduce the dependence on a lucky seed. A residual policy on top of the heuristic would keep its open-field competence and let learning handle splits. A reward term or a sub-goal for passing through openings could address the corridor and gate maps. Finally, the clone should be retrained on demonstrations from the procedural distribution before any claim is made about cloning as a method.

Limitations. Three seeds show the variance but are too few to estimate it well. The environment is a 2-D kinematic simulation with rectangular obstacles, and sight is not blocked by obstacles. The five presets are single maps and are useful as case studies, not as a distribution. The heuristic’s parameters were not retuned for the presets. We control a single herder; cooperation between herders [10], [11] is out of scope.

VIII. CONCLUSION

We trained a recurrent PPO herder under limited visibility on procedurally generated layouts with an adaptive curriculum, and evaluated it against a collect-and-drive heuristic and a behavioral clone on 150 held-out episodes per scenario. Compared with a policy trained on fixed layouts, mean success over three seeds on held-out split-field and open-field maps rose from 0% to 28% and 21%, and to 45% and 49% for the best seed. Two of three seeds beat the heuristic on the split field, and all seeds trail it on the procedural test suite and in open space. Scores on the procedural test were consistent across seeds, while scores on single maps were not. Getting the curriculum to work required four fixes: random goals from the first stage, a stage lock with a demotion margin, collision checks used only for promotion, and checkpoints saved with their normalization statistics. The next steps are more seeds, checkpoint selection that is less sensitive to the seed, longer training at full difficulty, and a reward for moving the flock through openings.

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